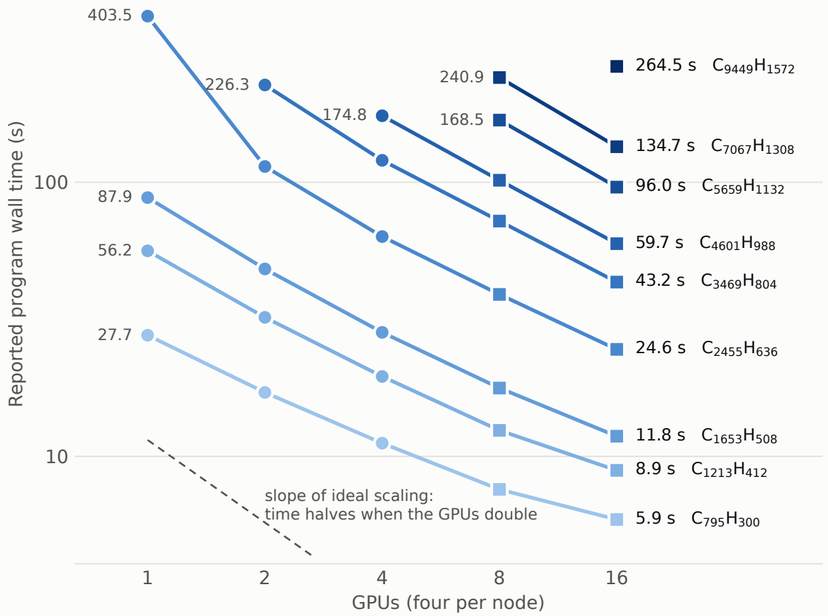
Complete SCF calculations on 1 to 16 GPUs: time, GPU memory and host memory

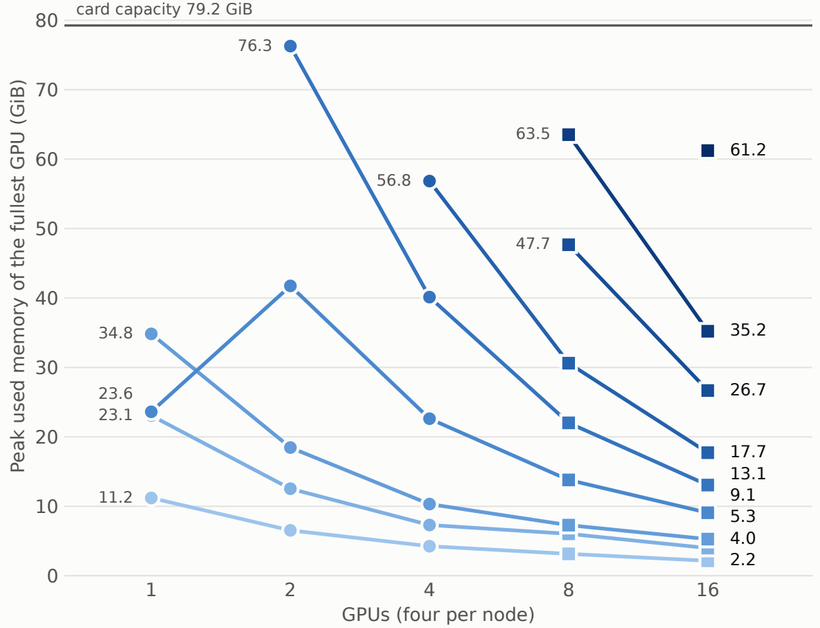
$C_{795}H_{300}$ (3,480 e) · $C_{1213}H_{412}$ (5,264 e) · $C_{1653}H_{508}$ (7,120 e) · $C_{2455}H_{636}$ (10,456 e) · $C_{3469}H_{804}$ (14,680 e) · $C_{4601}H_{988}$ (19,392 e) · $C_{5659}H_{1132}$ (23,768 e) · $C_{7067}H_{1308}$ (29,576 e) · $C_{9449}H_{1572}$ (39,368 e)

● one array per sector ■ basis shared by the GPUs of a sector

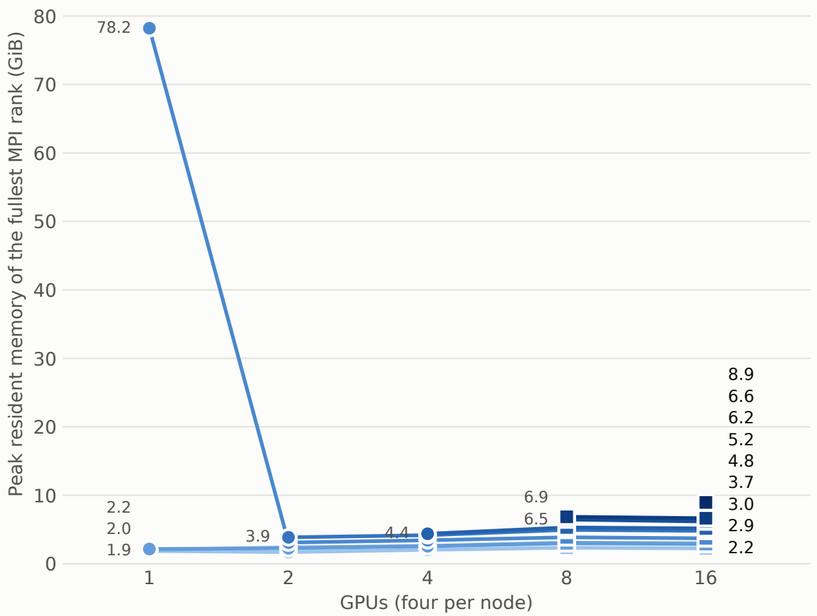
Complete SCF time



GPU memory



Host memory (one rank per node)



One measured run per point after a warm-up (C2455H636 on 1 GPU, C3469H804 on 2 GPUs: one run); totals within 1.7e-7 Ry of series F9, same SCF steps. GPU memory: nvidia-smi samples every 200 ms, fullest GPU, including contexts and allocator caches. Host memory: resident high-water mark of the fullest MPI rank. Series F10 (5 A of vacuum): code defaults, double precision, first calculations without symmetry cache. Source f8f3a50.